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Build an AI Second Brain with Claude Code



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<https://medium.com/@tahirbalarabe2/how-to-m...>



Introducing Today's Project!

In this project, I'm going to build a second brain using Claude Code and Obsidian to run oof the Andre Kapathy methodology. I will be learning about wikis, indexing and different queries to create my wn personal knowledge base for an LLM.

Key tools and concepts

Obsidian: A free note-taking app that uses plain Markdown files and provides a visual Graph View to see connections between ideas. Claude Code: A command-line AI agent that can read, create, and edit files directly on your computer, acting as the "wiki maintainer." Obsidian Web Clipper (Optional): A browser extension for saving web pages directly into the vault. Concepts: The Vault Architecture: The fundamental principle of separating a raw/ folder (your untouched source material) from a wiki/ folder (AI-compiled knowledge). This is called the "load-bearing rule" because keeping them separate ensures your original sources are never modified. LLM Wiki Pattern: Andrej Karpathy's methodology of using an LLM as a "programmer" and Obsidian as the "IDE" for a knowledge base. Schema-Driven Behavior: How a CLAUDE.md file acts as a constitution that the AI reads every session, defining project structure, page conventions. Slash Commands as Primitives: Knowledge Synthesis:

Challenges and wins

The project is estimated to take 60 minutes. From the tutorial, you would have spent this time on: Installing and setting up both Obsidian and Claude Code. Scaffolding the raw/ and wiki/ vault architecture. Defining your CLAUDE.md schema and slash commands. Exporting your existing data from other apps and converting it to Markdown. Running your first /ingest cycle and exploring the compiled wiki and Graph View.



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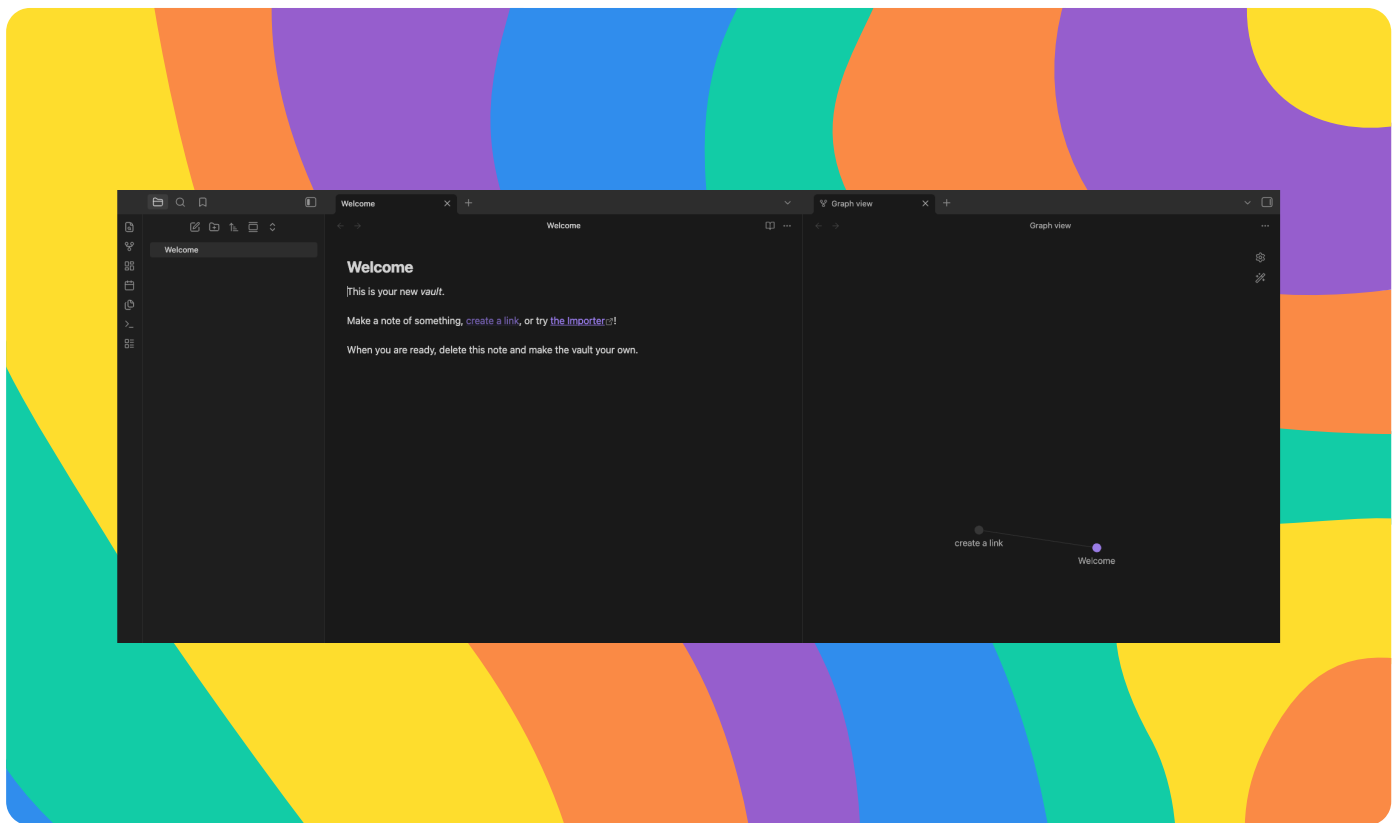
Why I did this project

I did this project today to learn how to build an AI Second Brain with Claude Code Another skill I want to learn is automating the ingest so I don't have to move files manually.



Installing Obsidian and Claude Code

In this step, I'm installing Obsidian and creating our vault, and then I will install Claude code to then sort things out later on in the project.



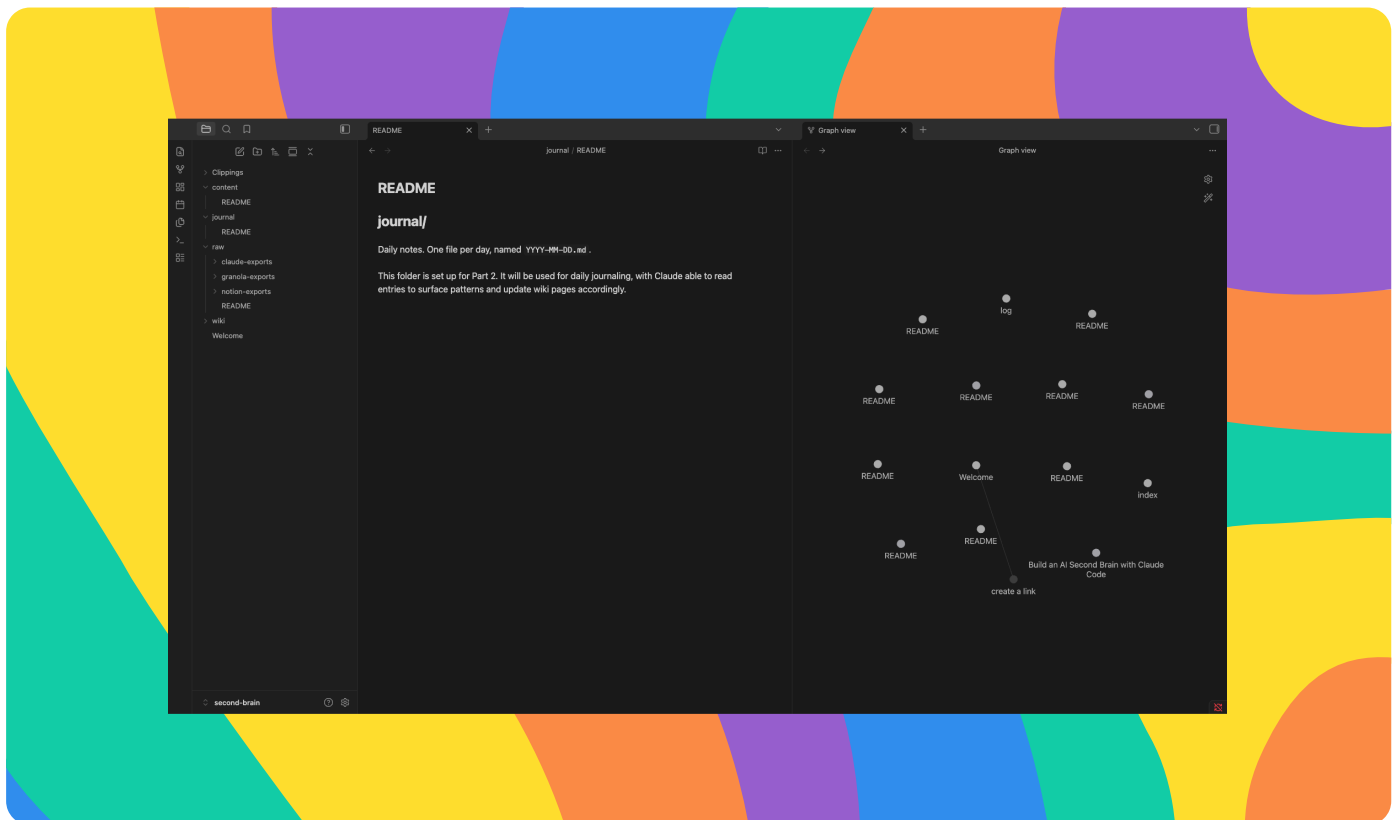
Verifying my Claude Code install

I installed Claude Code version 2.1.158



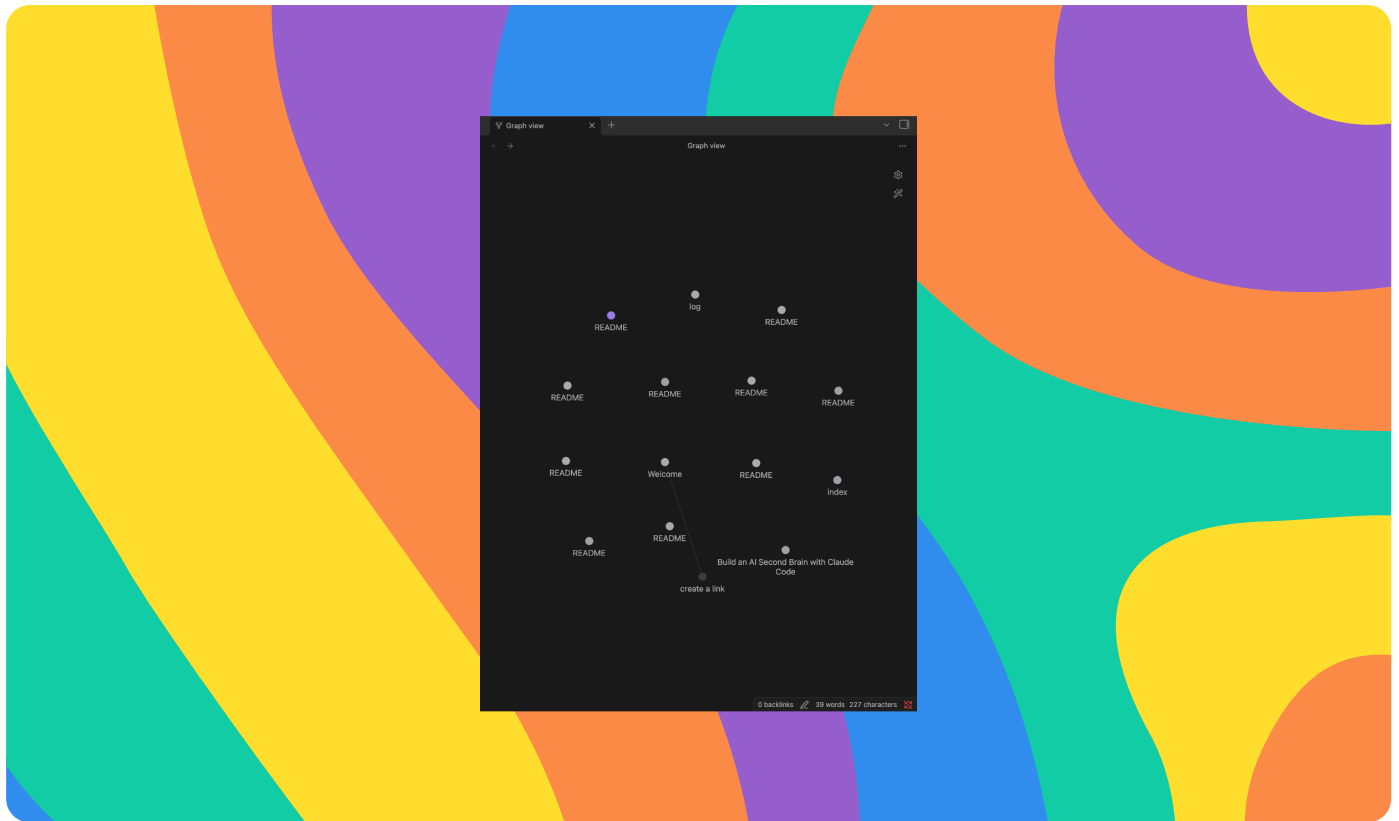
Building the Vault Architecture

In this step I am going to get Claude code to scaffold our vault's folder structure, and we will also be verifying the new folders appear in Obsidian. We are in particular going to pay attention to the raw and the wiki folders.



Inside the wiki/ folder

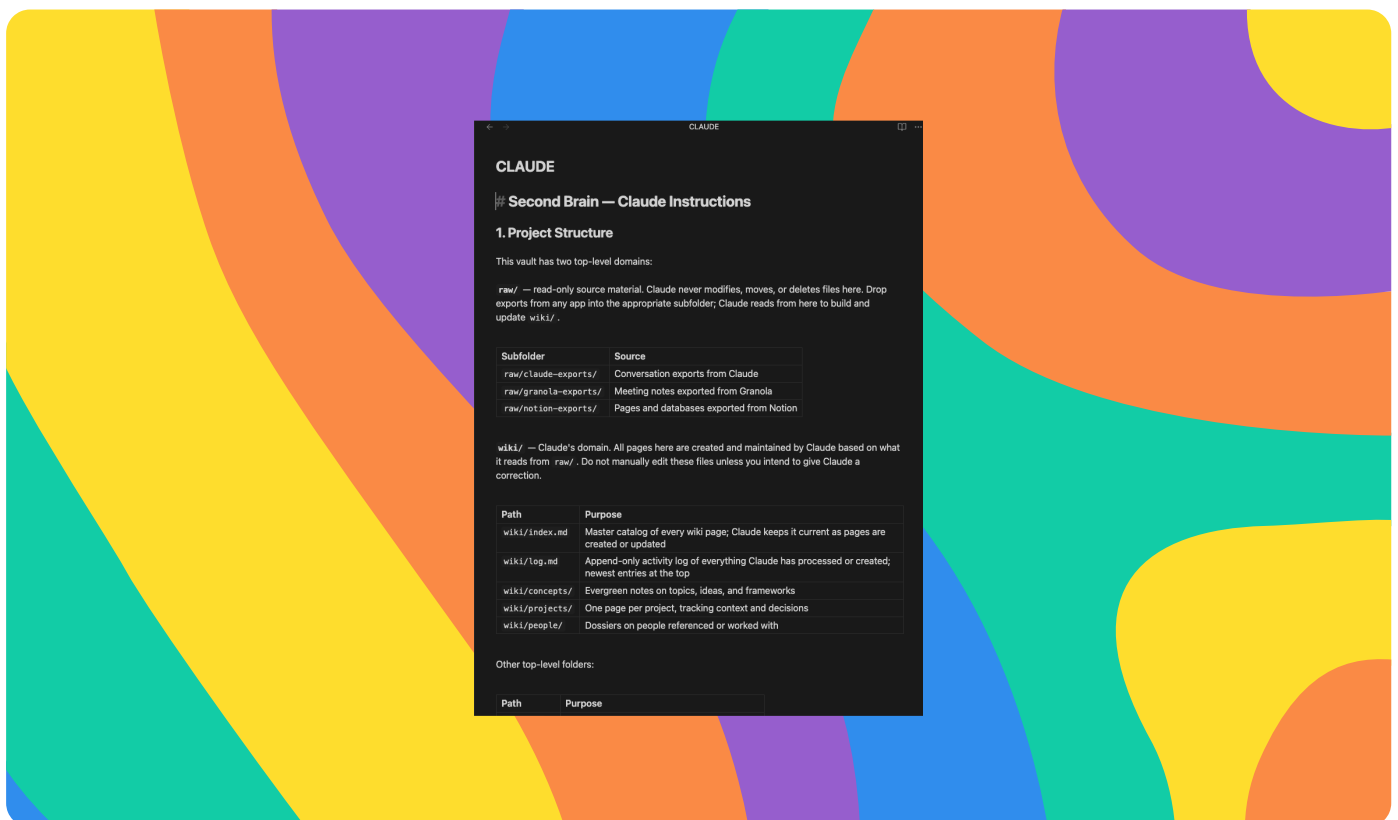
I counted 6 subfolders(well, it was actually four subfolders and two files).





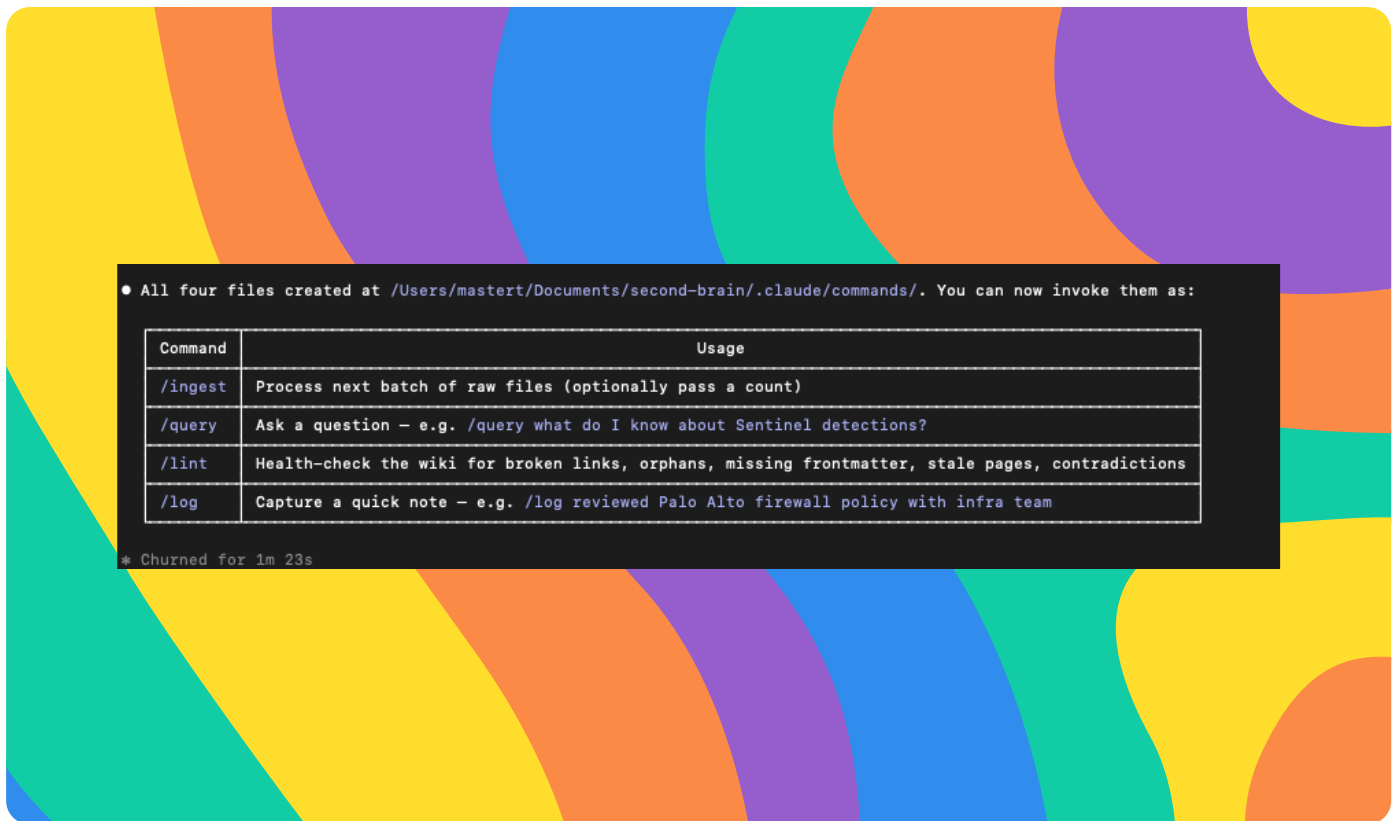
Configuring CLAUDE.md and Slash Commands

In this step, I'm creating a CLAUDE.md file to load into context every single time that i open a claude session within this project, it will understand: -the project structure -the domain context -the style guide -the page conventions



The four slash command workflows

The four sections in my CLAUDE.md are ingest, query, log, and lint. Each one instructs claude to perform different operations that i will cover later on in this project. In the screenshot, we actually can't see our claude commands as they are in a hidden claude folder.



Scaffolding the slash commands

`.claude/commands/` are real Claude Code slash commands: typing `/ingest` at the start of a message runs the file as a prompt, with `$ARGUMENTS` filled in from anything you type after the command. They tab-complete, they're shareable via Git, and they don't conflict with Claude Code's parser.



Available Slash Commands

Command	Purpose
/ingest	Ingest new files from raw/ into wiki/
/lint	Run a health check on the wiki
/log	Append a timestamped note to wiki/log.md
/query	Synthesize an answer from the wiki
/update-config	Configure the Claude Code harness via settings.json
/keybindings-help	Customize keyboard shortcuts
/verify	Run the app and observe behavior to confirm a change works
/code-review	Review the current diff for bugs and cleanups
/simplify	Review changed code for reuse, simplification, and efficiency
/fewer-permission-prompts	Scan transcripts and add allowlist entries to reduce permission prompts
/loop	Run a prompt or command on a recurring interval
/schedule	Create or manage scheduled remote agents on a cron schedule
/claude-api	Build, debug, and optimize Claude API / Anthropic SDK apps
/run	Launch and drive the project app to see a change working
/init	Initialize a new CLAUDE.md file with codabase documentation
/review	Review a pull request
/security-review	Complete a security review of pending changes

CLAUDE.md Rules

Section 1 - Project Structure. The vault has two top-level domains: raw/ is read-only source material (Claude never modifies, moves, or deletes anything here) organized into subfolders by source app; and wiki/ is Claude's domain, where all pages are created and maintained by Claude from what it reads in raw/. The wiki has a fixed structure: wiki/index.md as a master catalog, wiki/log.md as an append-only activity log, and subdirectories for concepts/, projects/, and people/. Other top-level folders (journal/, content/, cloudops/) exist outside both domains.

Section 2 - Page Conventions. Every wiki page must open with YAML frontmatter containing title, type (one of concept/entity/source-summary/comparison/project/person), sources, related, created, and last-updated. Internal links use [[page-name]] wiki-link syntax - never bare URLs. Pages must be atomic (one idea, person, or project per page; split if covering two distinct things), and headings must follow a strict hierarchy (# title, ## major sections, ### subsections, no skipping levels). After writing or updating any page, Claude must update wiki/index.md and append an entry to wiki/log.md.

Section 3 - Style Guide. Write in clear, concise prose - short paragraphs or bullets over long blocks. Every claim must be attributed to its source using inline citations or a ## Sources section. Contradictions between sources must be called out explicitly with a >=Contradiction= blockquote rather than silently resolved. Avoid filler phrases; state things directly. Use plain language and define jargon on first use. Do not add information not present in the source material - flag gaps with [needs source] rather than inferring.

Section 4 - Domain Context. The vault supports work at the intersection of cloud security, SecOps, and AI-driven automation for support operations; treat Palo Alto, AWS, Sentinel, Splunk, and related internal tooling as first-class entities worth their own wiki pages. The underlying goal is building security judgment; so prefer pages that capture reasoning behind decisions over pages that simply list facts. Two additional themes to flag: the transition from support work to systems ownership (flag concepts and projects that represent that shift), and AI systems that behave like operators rather than domes (when summarizing AI material, prioritize operational reliability and failure modes over capabilities and benchmarks).



Converting Source Notes Into the raw/ Folder

In this step, I'm going to what in my notes and conversations from different tools? Convert the JSON exports into markdown because that's the only format that Obsidian can read, and then we're going to verify files in Obsidian.



How many files Claude Code converted

I converted 42 conversations from Claude.ai. The script created 42 markdown files which were converted from JSON



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Checking the converted files in Obsidian

I verified that the converted files have YAML front matter. The frontmatter includes source, date, and topic.



Running the First Ingest Cycle

In this step I am to create or run the `/ingest` command to process our raw sources. I'm going to explore the wiki pages that Claude Code has compiled, and i'll be able to view this all in my obsidian knowledge graph.



```
• Done. Here's the run summary:

3 new wiki pages:
- [[enterprise-data-trust-architecture]] - the 7-layer PII architecture that emerged from the Palantir/NHS discussion, including the full Metadata Tax breakdown, vendor landscape, and the Layer 0 substrate concept
- [[world-labs-marble]] - Fei-Fei Li's generative 3D platform; the shift from symbol manipulation to spatial intelligence
- [[effective-prompting]] - Anthropic's official prompting principles distilled into a reference page, linked to [[risen-framework]]

4 updated pages:
- [[palantir-nhs]] - added accurate controversy details (IP ownership, redacted contract, 2026 political status, break clause review)
- [[apollo-research]] - added the precise scheming threat model (deception, sandbagging, evaluation manipulation) in Tahir's own words from the conversation
- [[risen-framework]] - added the agent skills blog application as a third applied example
- [[securing-ai-agents-data-governance]] - added the identified gaps in the source article (parameter injection, output governance, thin red-team, vendor lock-in)

16 conversation files remain unprocessed. Run /ingest to continue.
```

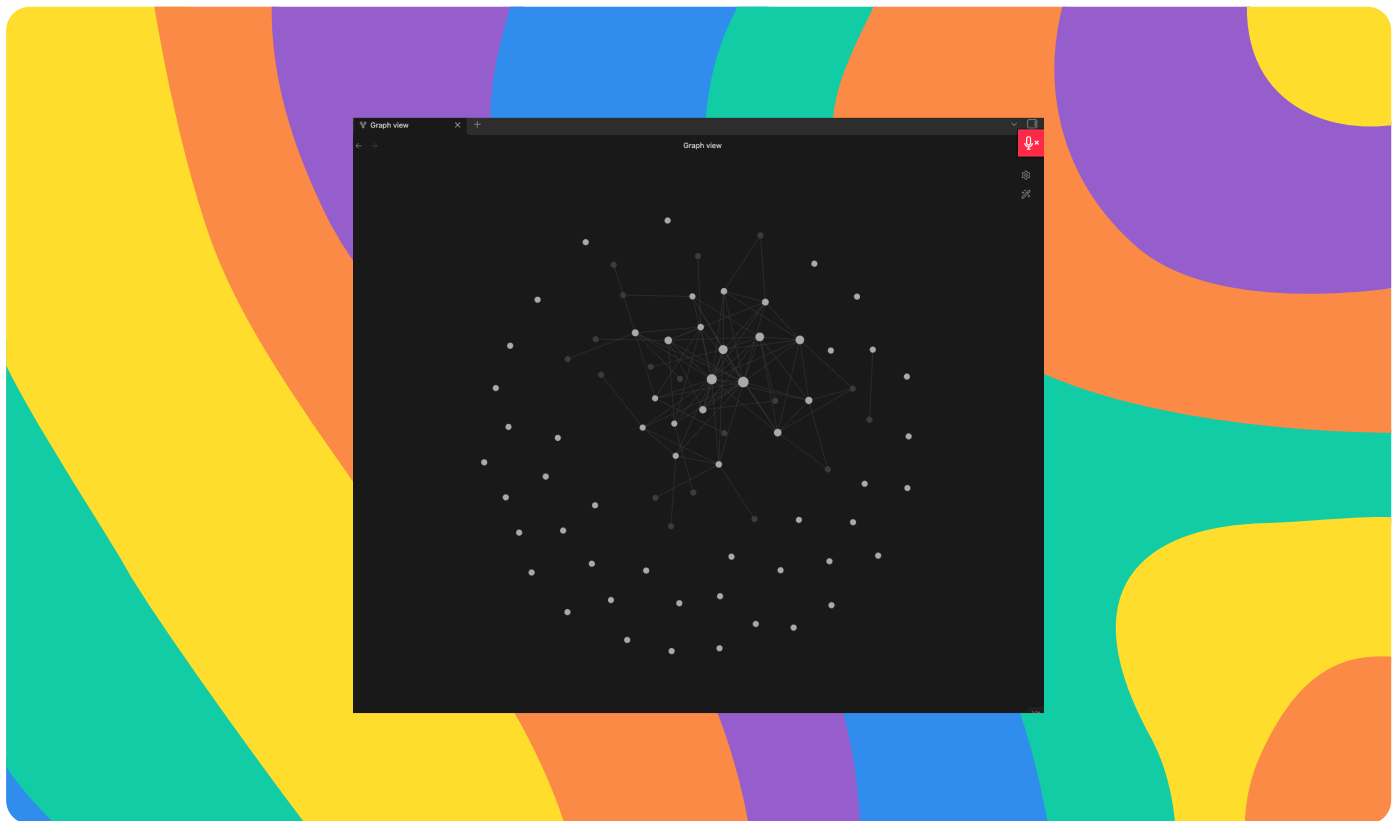
The pages Claude Code created

After the ingest, `index.md` lists the different connections, concepts, and tags so that we can see how these ideas get linked to other ideas.



Exploring a wiki page

I explored the page mobility page and it told me that i was focusing [[threat-modelling]], [[maestro-framework]], [[ai-agent-governance]], [[securing-ai-agents-data-governance]], [[risen-framework]], [[ics-security]], [[three-tier-web-architecture]],





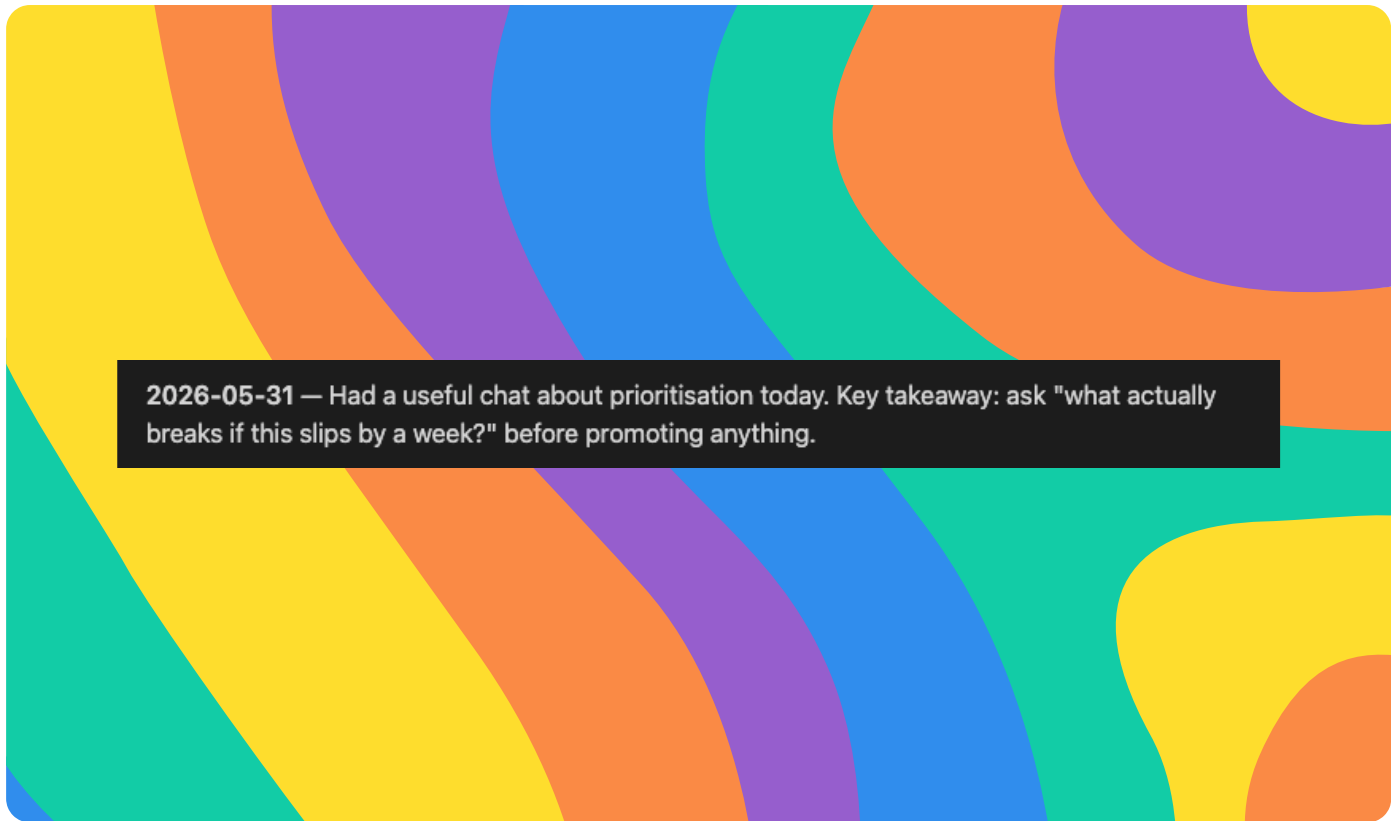
Secret Mission: Dogfooding /lint and /log

In this project extension, I'm going to test the /query/lint and /log slash commands.



What /lint found and how Claude fixed it

In this project extension, /lint reported I asked Claude to fix The three inaccurate last-updated dates are trivial one-liners. The 17 broken links need a decision first: should Sources sections use plain text instead of `[[[]]]` syntax, or is `[[raw-slug]]` acceptable as a citation convention?



2026-05-31 — Had a useful chat about prioritisation today. Key takeaway: ask "what actually breaks if this slips by a week?" before promoting anything.

Capturing a thought with /log

In this project extension, I captured a thought with /log about nothing No entity matches in the note. Claude also updated nothing as No wiki pages touched — no entity matches found.



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